

Python Fundamentals: From “What Does This Mean?” to Research

Before We Add Anything New: What Is Python?

- Python is a language for giving a computer instructions.
- You do NOT need to memorize everything.
- Our goal: look at simple code and understand what it is asking the computer to do.

Think: Python = instructions for the computer

What Is Google Colab?

- Google Colab is the place where we write and run Python.
- Python = the language.
- Colab = the notebook where we use the language.

Like: English = language • Google Docs = place to write

What Is a Code Cell?

- A code cell is a little box where we put an instruction.
- Press ► Run to ask the computer to follow that instruction.

```
print("Hello!")
```

Write instruction → Run → See what happened

The Most Important Rule: Read Code Left to Right

- Do not need to memorize computer code. Translate it into everyday language.
- Ask: What information do I have? What am I asking Python to do?

```
print(name)

name = "Alex"
```

Read it as: “Call this Alex” → “Show me Alex”

Quotation Marks " " Mean: This Is Text

- Words inside quotation marks are text.
- Without quotation marks, Python may think the word is a name for something else.

```
headline = "Gamecocks win season opener"
```

"Gamecocks win season opener" = text

= Means: Give Something a Name

- In Python, = usually does NOT mean a math equation.
- It means: store this information under this name.

```
Alex_screentime = 120
```

likes = 120 → “Remember 120 using the name likes.”

== Means: Ask Whether Two Things Are Equal (AKA Are True)

- One = stores information.
- Two == asks a True/False question.

```
Alex_screentime = 120 is not the same with  
Alex_screentime == 120
```

= → STORE | == → ASK

The Dot . Means: Use a Tool That Belongs to This Thing

- Start with the thing you have.
- The dot means: now use one of its tools.

```
headline = "BIG NEWS TODAY"  
headline.lower()
```

headline.lower() → “Take this headline and make it lowercase.”

Parentheses () Usually Mean: Do Something

- A function is an action.
- The parentheses hold the information the action needs.

```
print("Hello")  
len("journalism")
```

print(...) = show it • len(...) = count it

Square Brackets [] Usually Mean: A Group — or Pick From a Group

- We can put several things together in a list.
- Later, we can use [] to pick one thing from that group.

```
News_topics = ["sports", "AI", "music"]
```

[] = think “a group of things”

Means: A Note for Humans

- Python ignores anything after # on that line.
- Comments help us remember what the code is supposed to do.

```
# Save the headline  
headline = "AI changes journalism"
```

= explanation for you, not an instruction for Python

Quick Check: Can You Read This?

```
                # Our example  
headline = "AI Changes Journalism"  
clean_headline = headline.lower()  
print(clean_headline)
```

If you can explain the code in words, you are learning Python.

What We Know So Far

- Text → quotation marks
- = → store / give a value a name
- == → ask whether two things are equal
- . → use a tool that belongs to something
- () → do an action / give an action information
- [] → group things or pick something
- # → note for humans

Today we will use these same pieces — not start over.

Today's Research Story

Imagine you collected 100 headlines. What could Python help you do?

The Research Problem Comes First

- Question: How often do news headlines mention AI?
- You collect many headlines.
- Now you need to store them, look through them, and count what you find.

Research question first → code second

1. Store Many Headlines

New idea: a list

A List Is Just a Group of Things

- Think of a playlist: one playlist contains many songs.
- A Python list can contain many headlines.

```
headlines = [  
    "AI changes journalism",  
    "Gamecocks win opener",  
    "New music festival announced"  
]
```

One variable can hold MANY pieces of information.

Why Would a Researcher Need a List?

- 1 survey response → one piece of information.
- 500 survey responses → a collection.
- 1 headline → one piece of information.
- 10,000 headlines → a collection.

Research usually means working with MANY things.

2. Find One Thing in the Group

New idea: an index

Python Gives Each Item a Position Number

- Python starts counting at 0. This feels strange at first.
- 0 = first item • 1 = second item • 2 = third item

```
headlines[0]
```

headlines[0] means: “Give me the FIRST headline.”

Try to Predict Before You Run

- What will each line return?

```
headlines[0]  
headlines[1]  
headlines[2]
```

Prediction first → then Run → then check yourself

3. Repeat the Same Job

New idea: a loop

Why Repeat Automatically?

- Imagine you have 100 headlines.
- You could print headline 1, then headline 2, then headline 3...
- Or tell Python: do the same thing for every headline.

A loop = repeat an instruction

A Loop, Translated Into English

```
for headline in headlines:  
    print(headline)
```

- Read the first line as: For each headline in my group of headlines...
- Read the second line as: ...show me that headline.

Take one → do the job → take the next → repeat

Why Is the Second Line Moved Over?

- The spaces are called indentation.
- Indented = this instruction belongs inside the loop.
- Python uses the spacing to understand what should repeat.

```
for headline in headlines:  
    print(headline)  
  
print("Finished")
```

Indented → repeat • Not indented → do after the loop

4. Ask Python a Yes/No Question

New idea: if

if Means: Only Do This When Something Is True

- Everyday example: IF it is raining, take an umbrella.
- Research example: IF a headline mentions AI, mark it.

```
headline = "AI changes journalism"
```

```
if "AI" in headline:  
    print("mentions AI")
```

if = check a condition before doing something

Now the Pieces Start Working Together

- We already know: list = many headlines.
- Loop = look at each headline.
- if = check whether each headline contains AI.

```
for headline in headlines:  
    if "AI" in headline:  
        print(headline)
```

This is the beginning of a real text-analysis workflow.

What Is the Researcher Doing Here?

- The researcher decided the question.
- The researcher decided what counts as “mentions AI.”
- Python is doing the repetitive checking.
- You still inspect whether the output makes sense.

Python does the repetition. You do the research thinking.

5. Let AI Help — Without Giving Up Understanding

You can ask an LLM for code, but you should be able to explain the pieces.

A Better Way to Ask AI for Coding Help

- Tell AI your research goal.
- Ask for the simplest possible code.
- Ask it to explain every line in plain English.
- Run a tiny example first.
- Check whether the output is what you expected.

AI-generated code = proposed solution, not verified solution

Your Goal Is NOT to Memorize Python

- By the end of this course, I want you to be able to:

Explain what simple code is doing.

Use AI to help write or fix code.

Notice when the output does not make sense.

Connect code back to a research question.

Understanding > memorizing syntax

Assignment 2: Start in Next Thursday's Lab

Human + AI Literature Screening



1. Choose a Research Question

SPORTS JOURNALISM / SPORTS MEDIA

- How does social media change the way sports journalists report breaking news?
- How does social media coverage influence fans' perceptions of athletes?
- How does sports media coverage differ between men's and women's sports?

SOCIAL MEDIA & MEDIA EFFECTS

- How does short-form video affect how people consume news?
- How does social media use affect trust in news and information?
- How does social media content influence people's perceptions of public figures?



1. Choose a Research Question

STRATEGIC COMMUNICATION / MARKETING

- How does influencer marketing affect consumers' trust in brands?
- How does a company's response to a social media crisis affect public opinion?
- How does AI-generated advertising affect perceptions of brand authenticity?

MEDIA, SOCIETY & PROFESSIONAL ISSUES

- How does news coverage influence public perceptions of high-profile court cases?
- How does media literacy affect people's ability to identify misinformation?
- How does generative AI affect students' research and writing practices?

2. Prepare Your Abstract Set



Groups of 4-5



Choose one research question



Find **3 academic abstracts** PER PERSON



Resources: Google Scholar. Select Only Peer-Reviewed Articles



Save as one .csv file



Bring the file to the Week 4 lab

What Are Peer-Reviewed Articles?

- Research articles evaluated by other experts before publication
- Usually published in academic or scholarly journals
- Look for articles that include an **abstract, methods, results, and references**
- Find them through **Google Scholar** or the **USC Libraries databases**
- For this activity, use **peer-reviewed journal articles only**
- Avoid news articles, blogs, websites, reports, theses, and book chapters

3. Save Your Abstracts Like This (One File Per Group)

Your CSV should contain:

Group member	abstract
Alex	This study examines how journalists...
Alex	Using survey data, this study explores...
Alex	This paper investigates physicians...
Sara	...
Sara	...
Sara	...

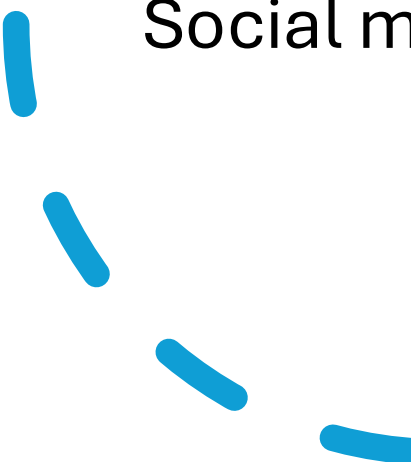
4. Your Abstract Set Should Be Mixed

Try to include:

- **One clearly relevant:** Directly addresses your question
- **One possibly relevant:** Related construct, population, or technology
- **One probably unrelated:** Similar keywords, different research problem



Example question: **How does social media coverage influence fans' perceptions of athletes?**

- **Clearly relevant:**
Social media coverage of athletes + fan perceptions
 - **Possibly relevant:**
Athletes' own social media use + fan engagement
 - **Probably unrelated:**
Social media advertising + sporting event attendance
- 

Next Class: ZOOM LAB



Zoom link for this Thursday
is posted on Blackboard.



Do not come to class next
Thursday

